

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Summer Examination – 2024 Course: B. Tech. Semester :VII Branch: Electronics and Communication Engineering (Sandwich) / Electronics and Communication Engineering / Electronics and Telecommunication Engineering Subject Code & Name: BTEXC701 / BTETC701 Microwave Engineering Max Marks: 60 Date: 01/07/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/ CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Discriminate Transmission line and Waveguide.	CO1	6
B)	Derive the general solution for TEM/TE/TM wave.	CO1	6
C)	Write note on Cavity resonator in Circular Waveguide.	CO1	6
Q.2	Solve Any Two of the following.		12
A)	Explain Magic Tee with help of S matrix.	CO2	6
B)	Explain working principal of Isolator and Circulator with help on neat and labeled diagram.	CO2	6
C)	Explain application Directional coupler in detail.	CO2	6
Q. 3	Solve Any Two of the following.		12
A)	What are the limitations of Conventional tubes.	CO3	6
B)	Explain two cavity klystron with help of velocity modulation and Applegate diagram and Bunching process.	CO3	6
C)	Explain Construction and Principle of operation of 8 cavity cylindrical travelling wave magnetron,	CO3	6
Q.4	Solve Any Two of the following.		12
A)	Explain VSWR meter and Power Meter with neat labeled diagram.	CO4	6
B)	Explain Attenuation measurement, Phase shift measurement with help of block diagram.	CO4	6
C)	Explain Impedance measurement, Q of cavity resonator measurement with	CO4	6

	help of block diagram.		
Q. 5	Solve Any Two of the following.		12
A)	Explain Structural details and applications of Stripline.	CO5	6
B)	What are the different types of Stripline explain in detail?	CO5	6
C)	What are the hazards of Electromagnetic Radiation on the human body?	CO5	6
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